

Esteban Sitek

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **12+** years across fintech, healthcare, and e-commerce, architecting enterprise-scale data platforms using **Databricks** and **PySpark** with strong emphasis on data quality, testing, and modern cloud infrastructure. Proven expertise in migrating legacy systems, building batch and streaming pipelines, and establishing CI/CD best practices for analytics-driven organizations.

SKILLS

Languages: Python, SQL, Scala, JavaScript, TypeScript, HTML, CSS

Data & ML: Databricks, PySpark, Delta Lake, Spark SQL, Apache Spark, Airflow, dbt, Pandas, NumPy, Medallion Architecture

Database: PostgreSQL, MongoDB, Redis, Elasticsearch, Unity Catalog

Testing: pytest, Great Expectations, dbt tests, Integration Testing, Data Quality Validation, Regression Testing

Cloud & DevOps: Azure, AWS, GitHub Actions, Jenkins, Docker, Kubernetes, Terraform, Databricks Workflows

Others: RESTful API, Git, Jira, Confluence, Data Lineage, Data Governance, Agile

EXPERIENCE

Publicis Sapient

Jun 2024–Present

Senior Software Engineer

- Led the design and implementation of enterprise data pipeline modernization initiatives using **Databricks** and **PySpark**, handling complex transformation logic across financial services platforms and delivering 40+ batch jobs with zero data loss.
- Architected a multi-stage **Medallion Architecture** implementation (bronze, silver, gold layers) using **Delta Lake** and **Unity Catalog**, enabling business analysts and data scientists to access certified datasets while maintaining strict governance controls across 200+ downstream consumers.
- Established comprehensive data quality frameworks using **Great Expectations** and custom **PySpark** validators, implementing automated reconciliation processes that reduced production data incidents by **68%** and eliminated manual validation overhead.
- Designed and deployed CI/CD pipelines using **GitHub Actions** and **Terraform**, automating Databricks job deployments and enabling safe promotion from dev through staging to production with integrated **pytest** and integration testing.

- Collaborated with cloud architecture teams on Azure migration strategy, migrating legacy batch systems from on-premises infrastructure to managed **Databricks** clusters, reducing compute costs by **52%** while improving query performance.
- Mentored three junior data engineers on PySpark best practices, code review standards, and **Databricks** workspace optimization, establishing reusable templates and documentation that accelerated team onboarding by 6 weeks.
- Optimized existing **Apache Spark** jobs through careful partitioning strategy and adaptive query execution, reducing job duration from 4 hours to 18 minutes on average cluster runs.
- Implemented streaming ingestion framework using **Spark Structured Streaming** for real-time financial transaction feeds, processing 2.5M+ events daily with sub-second latency and exactly-once delivery semantics.
- Developed automated testing suite combining **pytest** unit tests with **dbt** data transformation tests and **Great Expectations** data contracts, reducing regression cycle time by **45%**.
- Performed troubleshooting on failed pipeline executions across Databricks jobs, diagnosing performance bottlenecks, optimizer issues, and data schema mismatches to restore SLAs within 2–4 hours.

CareSimple

Oct 2022–May 2024

Senior Full Stack Engineer

- Spearheaded the redesign of healthcare data ingestion pipelines using **PySpark** and **Spark SQL**, consolidating 12 disparate ETL jobs into a unified framework that processed 180+ data sources across hospital systems, reducing infrastructure costs by **38%**.
- Built batch and streaming data processing systems for remote patient monitoring (RPM) data using **Databricks** and **Delta Lake**, ensuring HIPAA compliance through encryption, access controls, and immutable audit logs that satisfied healthcare compliance audits.
- Implemented **Medallion Architecture** principles across the analytics platform, organizing raw (bronze), cleaned (silver), and aggregated (gold) layers using **Spark SQL** and **PySpark** transformations that made healthcare datasets queryable for 20+ downstream BI dashboards.
- Established data quality validation pipelines using **Great Expectations** integrated with **Airflow**, catching upstream data anomalies before they propagated downstream, reducing alert noise by **55%** and improving data reliability for clinical teams.
- Designed and maintained automated testing framework combining **pytest** for transformation logic and **dbt** for SQL model validation, covering 85% of transformation codebase and catching regressions before production deployment.
- Collaborated with platform team to implement **GitHub Actions**-based CI/CD for data assets, enabling safe code reviews and automated promotion of PySpark jobs across dev, staging, and production environments.
- Troubleshot complex data inconsistencies in patient datasets across multiple hospital systems, using **Spark SQL** and **Delta Lake** time-travel features to trace data lineage and recover corrupted records within 8 hours.

- Mentored two junior engineers on **PySpark** optimization techniques and **Databricks** workspace best practices, resulting in improved query performance across team-owned jobs by average **32%**.

Shopee

Jul 2018–Sep 2022

Full Stack Engineer

- Architected batch data processing pipelines using **Apache Spark** and **PySpark** for e-commerce analytics, ingesting order, inventory, and seller data from 15+ Southeast Asian markets and serving 200+ analytics dashboards.
- Implemented **Medallion Architecture** design patterns across the data lake, organizing transaction data into structured bronze, silver, and gold layers using **Spark SQL** transformations that enabled self-service analytics for 50+ business stakeholders.
- Built real-time data streaming framework using **Spark Structured Streaming** for inventory synchronization across Shopee's marketplace platform, processing 800K+ events per second with 99.95% reliability SLA.
- Established data quality framework using custom **PySpark** validators and integration tests, reducing downstream analytics incidents by **41%** and enabling confident data consumption across all product teams.
- Deployed CI/CD pipeline for data assets using **Jenkins** and **Terraform**, automating **PySpark** job deployments and schema validation, cutting deployment time from 3 hours to 22 minutes.
- Optimized legacy batch jobs through careful **Spark** partitioning, caching strategies, and dynamic resource allocation, reducing nightly ETL duration from 6 hours to 1.5 hours and freeing compute resources for interactive queries.
- Collaborated with data science team to build feature engineering pipelines using **PySpark** and **Pandas**, preparing ML-ready datasets for real-time product recommendation engine serving 10M+ daily requests.
- Performed production incident response on failed data pipelines, diagnosing data corruption issues and implementing automated reconciliation checks in **PySpark** to prevent silent failures.

Altitude Games

Nov 2014–Jul 2018

Full Stack Engineer

- Built data analytics platform for mobile game telemetry using **Python** backend services and **MongoDB** data store, collecting 50M+ daily player events and powering real-time dashboards for game balancing.
- Implemented batch ETL jobs using **Python** to aggregate player behavior data, feeding insights to product managers on retention, monetization, and feature engagement across five published titles.
- Developed REST API endpoints in **JavaScript** and **Python** that exposed analytics data to internal tools, enabling designers and business teams to query player demographics and spending patterns.
- Established automated testing for data ingestion pipelines using **pytest**, validating event schema compliance and catching corrupted telemetry before it reached production databases.
- Collaborated on infrastructure deployment using **Docker** containers and **GitHub** version control, enabling consistent environments across dev and production for game backend services and analytics jobs.

EDUCATION

Ateneo de Manila University
Bachelor of Science in Computer Science

Aug 2010–Jul 2014